



MTNet: Multimodal transformer network for mild depression detection through fusion of EEG and eye tracking

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ABSTRACT

With the increasing multimodality of biomedical data, data fusion has been widely utilized in depression research, in response to the significant rise in the incidence of depression. In our study, we propose a novel multimodal transformer network (MTNet) that integrates eye tracking with EEG data for depression detection. The MTNet effectively captures both local and global spatio-temporal characteristics of multimodal data, utilizing the global receptive field of the transformer model to capture the long-term dependencies within the data. Results confirmed that our proposed MTNet surpasses the feature-based fusion and other baseline models, with a classification accuracy of 91.79 %, which highlights the potential value of multimodal fusion with biological signals in depression detection. Moreover, we specifically concentrate on the fusion stage in multimodal fusion, carrying out analysis for early, intermediate, and late fusion respectively. Our study demonstrates that intermediate fusion yields superior performance in processing EEG and eye tracking data, effectively adapting to the heterogeneity of multimodal data.

1. Introduction

Depression, characterized by continuous and long-term low mood, has emerged as the most prevalent form of mental illness, affecting over 350 million individuals worldwide [1]. This condition ranks as the second leading cause of mortality [2], following heart disease, posing a significant threat to our well-being. Alarming, depression is increasingly affecting younger individuals, with a rising incidence among adolescents [3]. Beyond that, depression in adolescents is often overlooked in comparison to adults [4]. Given its relapsing nature, prompt detection and treatment of depression, particularly during middle to late adolescence, are of utmost importance.

At present, scale analysis and doctor-patient communication have become the predominant diagnostic methods for depression [5]. Individuals with depression may, for various reasons, conceal their true feelings during the diagnostic process, potentially leading to misdiagnosis. Moreover, discrepancies in professional experience among psychiatrists across different regions, along with variations in diagnostic standards for patients, further complicate the challenge of accurate

diagnosis. Consequently, an increasing number of studies are actively investigating biological indicators that can objectively reflect the level of depression, aiming to address these challenges [6–10]. Researchers have been focusing on investigating different bio-signals to detect depression, including electroencephalogram (EEG) [11–13], electromyography (EMG) [14], electro-oculogram (EOG) [15], and eye tracking (ET) [16,17]. Among these, EEG and eye tracking have gained popularity in psychological experiments due to their non-invasiveness and ease of recording.

Eye tracking is a valuable technique used in psychology for recording eye behavior, including gaze position, fixation time, and pupil dilation, which has proven to be useful in depression detection [18–20]. Stolicyn et al. [18] developed a depression assessment system that utilizes eye tracking technology to extract facial and eye tracking features, achieving a classification accuracy of 79 % using SVM. Likewise, Yang et al. [20] introduced a spatiotemporal clustering algorithm for depression detection, which directly uses eye tracking data to identify Regions of Interest (ROIs), achieving a classification accuracy of 76.25 % based on data from 89 participants.

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Compared to eye tracking, the acquisition rate of EEG devices is significantly higher. EEG is a non-invasive technology for diagnosing diseases, providing an objective indicator of mental and cognitive disorders due to its accurate reflection of human brain activity, especially for depression detection [21–24]. Mahato and Paul [23] employed a frequency band-based approach to extract features from the EEG data, they utilized multi-cluster feature Selection and achieved the accuracy of 88.33 % using SVM.

In contrast to machine learning algorithms that require manual feature extraction from EEG data above, deep learning excel in automatically extracting features for depression detection, showing superior performance. Long short-term memory (LSTM) and graph convolutional network (GCN) have been extensively utilized in research of EEG [25–27]. Notably, the emergence of transformer model [28] provides the potential to further enhance the classification performance of EEG data [29–32]. Yin et al. [30] presented a model that combines transformer and parallel convolutional neural network to extract valuable information efficiently, which outperforms existing strategies on two publicly available datasets. Besides, Tigga and Garg [32] introduced an attention-based transformer time-series model aimed at effectively detecting EEG perturbations in depressive patients, achieving an accuracy of up to 98.67 %.

With the increasing multimodality of biomedical data, there has been a rising application of data fusion across various fields [33–35]. Data fusion can be categorized into two categories: feature-based fusion and deep learning fusion, which depend on the need for manual feature extraction. When it comes to feature-based fusion, features are manually extracted from the multimodal data and then combined using machine learning algorithms in general. Deep learning fusion, on the other hand, leverages a multi-layer neural network to automatically extract features. Ding et al. [36] employed EEG, eye tracking, and galvanic skin response data to classify major depression, ultimately achieving an accuracy of 79.63 %. Similarly, Zhu et al. [37] presented a content-based ensemble method (CBEM) that combined EEG and eye tracking data, aiming to evaluate its effectiveness in identifying patients with depression. Moreover, the relationships between input biomedical data encompasses factors such as complementarity, redundancy, and collaboration. The fusion of EEG and eye tracking falls into the category of collaboration [38,39], as the combined information becomes more intricate and comprehensive than the original data sources.

The overall framework of our work is depicted in Fig. 1. When it comes to multimodal data fusion, our research focuses on the fusion strategy and fusion stage for EEG and eye tracking data. We

implemented two fusion strategies, feature-based fusion and deep learning fusion, which differ based on the requirement of manual feature extraction. The main work of the proposed study is listed as follows.

- 1) We developed a multimodal signal acquisition platform to investigate the visual attention mechanisms and cognitive brain activities of individuals with depression using non-invasive EEG and eye tracking technologies. This platform offers a comprehensive view to overcome the limitations of single-mode data in capturing physiological signal characteristics.
- 2) We propose a novel multimodal transformer network (MTNet) by integrating EEG and eye tracking data. The MTNet efficiently captures the spatio-temporal characteristics of multimodal data at both local and global scales. By leveraging the global receptive field of the transformer model, it effectively captures long-term dependencies within the data, underscoring the significant potential of multimodal fusion with biological signals in depression detection.
- 3) Our research delved into the impact of fusion strategies on multimodal data integration [40], examining early, intermediate, and late fusion. Our findings indicate that intermediate fusion exhibits superior performance in processing EEG and eye tracking data. This strategy effectively accommodates the heterogeneity of multimodal data and captures the interplay between modal features.

2. Materials and methods

2.1. Subjects and data acquisition

Our study included a total of 49 adolescents from Xi'an Gaoxing No. 1 High School, comprising 21 participants with mild depression (Mean age = 16.5, SD age = 0.46) and 28 Health control (HC) participants (Mean age = 16.7, SD age = 0.53). The exclusion criteria included instances where the eye tracking data accounted for less than 25 % or when EEG experiments were incomplete. The study was conducted in accordance with the principles outlined in the Declaration of Helsinki and received approval from the Ethics Committee of the Third Affiliated Hospital of Anhui Medical University (Ref.: 2020-020-01), located in Hefei, China. Written informed consent was obtained from all participants and their parents.

We used the DSM-V assessment scale by Professor Wang Jianping from Beijing Normal University to assess depression severity using the Level 1 Self-rated Symptom Scale and Level 2 Depression-Children Ages 11–17 Scale over the past 14 days [41]. Participants with mild

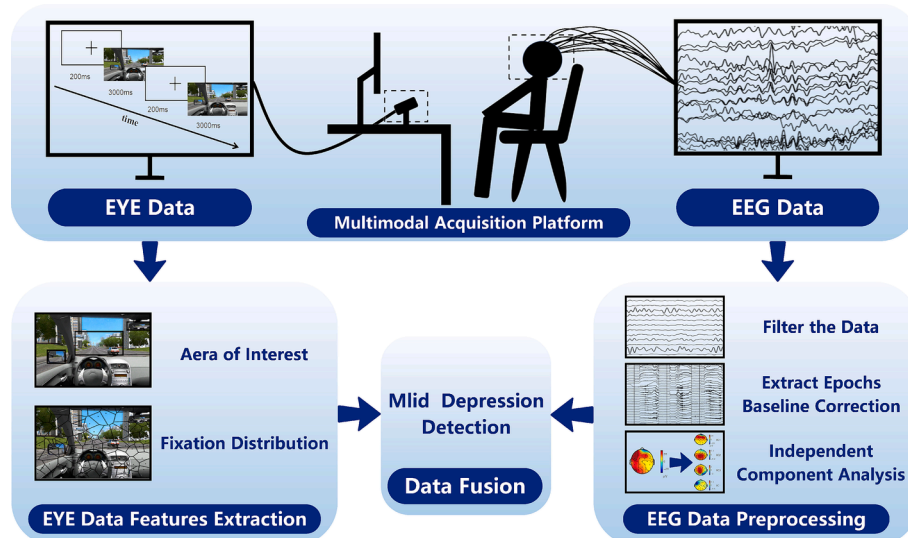


Fig. 1. The workflow of the proposed platform for depression detection.

depression scored within the range of 32 to 38. Our study employed an attention mechanism paradigm involving 20 trials with sudden passer-by appearances and vehicle collisions [42].

2.2. Data preprocessing

2.2.1. EEG Data preprocessing

The EEG signals were recorded using a 64-channel HydroCel Geodesic Sensor Net (Electric Geodesics, Inc.) at a sampling frequency of 500 Hz, referenced to Cz with electrodes impedance below 50 k. EEG data preprocessing was carried out in MATLAB (version 2017b) and EEGLAB [43] to address various noises and artifacts like blinking, head movement, and power frequency noise. Preprocessing steps included filtering (low-pass at 40 Hz, high-pass at 0.1 Hz), downsampling to 250 Hz, and extracting 20 four-second segments per subject, aligned around stimuli. Baseline correction involved subtracting the pre-stimulus mean, and channel interpolation was performed using the spherical interpolation algorithm in EEGLAB to handle bad channels. Additionally, Independent Component Analysis (ICA) was applied to remove eye and head movement artifacts.

2.2.2. EYE Data features extraction

The eye tracker utilized in our study is the EyeCatch Bar, developed by Dongguan Dongquan Technology Co., LTD. It provides an accuracy of 1.25 degrees and operates at a sampling rate of 41 Hz. Areas of interest (AOI) were defined to assess participants' attention mechanisms, including side mirrors, rearview mirrors, and front windscreens inside the car. Three types of eye tracking features were extracted based on the experimental paradigm: AOI sample percentage, AOI fixation time and fixation distribution [44]. Fixation distribution, a quantitative index derived from a Voronoi diagram, characterizes the clustering of discrete fixation points and explores the distribution of attention. A larger fixation distribution indicates more dispersed attention distribution.

2.3. Feature-based fusion

2.3.1. EEG features extraction

Feature extraction is a crucial tool in exploring brain function mechanisms and supporting computer-aided diagnosis. After EEG preprocessing, the main aim of feature extraction is to condense data into a reduced set, enabling efficient use of machine learning (ML) for classification and diagnostics. Given the non-stationary and nonlinear nature of EEG signals, our study employs statistical features (mean, kurtosis, skewness), Hjorth features [45] for activity quantification (activity, mobility, and complexity), and various nonlinear features like detrended fluctuation analysis (DFA), power spectral density, approximate entropy, and sample entropy to comprehensively analyze EEG signals.

2.3.2. Machine learning classifiers

Through feature extraction, the eye tracking and EEG data are compressed to 3 eye tracking features and 10 EEG features. The eye tracking data of a single participant, combined with 20 experimental trials and 3 eye tracking features, is compressed into a 60-dimensional eye tracking feature vector. Similarly, the EEG data, combined with 20 experimental trials and 10 EEG features, yields a 200-dimensional EEG feature vector.

Different machine learning algorithms may yield varying performance depending on the biological signal types and the size of the feature vector. Hence, we selected seven machine learning algorithms, which encompass support vector machine (SVM), adaboost, random Forest, k-nearest neighbors (KNN), naive bayes, and decision Tree. Utilizing these algorithms along with the extracted multimodal features, we performed classification experiments on the eye tracking data, EEG data, and the fusion data of them.

2.4. Deep learning fusion

2.4.1. The Architecture of multimodal transformer network

The structure of our proposed MTNet is depicted in Fig. 2. Inspired by previous research [46,47], we employed a fusion approach utilizing eye tracking and EEG data. We extracted EEG data specifically within the 4 Hz to 40 Hz range and segmented the frequencies at 4 Hz intervals, resulting in a total of 9 distinctive segments. We excluded the 0.1 Hz to 4 Hz band because this range corresponds to alpha waves, typically observed when individuals close their eyes or enter a state of sleep. Then we utilize a combination of spatial filter and temporal convolution layer to capture spatiotemporal characteristics from EEG data.

The temporal convolution layer has 40 kernels of size (1, 20) with a stride of (1, 1). To accelerate training and mitigate overfitting in temporal convolution layer, we incorporate batch normalization and employ the exponential linear unit (ELU) as the activation function. Additionally, we utilize average pooling after the activation function with the kernel size of (4, 60) and a stride of (1, 15), in order to reduce computational complexity, the processed data is subsequently fed into the Transformer Layer in the form of tokens.

In the upper right corner of Fig. 2, the Transformer Layer is depicted, consisting multi-head self-attention (MHSA) and feed forward network (FFN) modules. Each sub-layer is equipped with a residual connection and followed by layer normalization. MHSA is specifically utilized to capture global temporal dependencies in EEG data, effectively compensating for the limited receptive field of the convolution layer [28]. Additionally, the feed forward network comprises two linear projections, the gaussian error linear unit (GELU) activation function is applied between them.

The feature vector obtained from the temporal convolution layer are linearly transformed to generate three equivalent sets: query (Q), key (K), and value (V), with matching shapes. The correlation between Q and K is evaluated using dot products, incorporating a scaling factor to mitigate the issue of vanishing gradients. Subsequently, the output is passed through the softmax function and weighted by V using a dot product. d_k is the dimension of the key and d_{model} is the dimension of the output. The process can be expressed as follows; The entire attention computation is repeated N times.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

To incorporate information from different representation subspaces of each mode at various locations, we employ multiple attention functions to implement multi-head attention. The structure of our model is determined by the number of attention heads, denoted as h . The calculation of MHSA can be described as follows:

$$MultiHead(Q, K, V) = Concat(H_1, H_2, \dots, H_h)W^O \quad (2)$$

$$H_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (3)$$

where learnable projection matrices $W_i^Q \in R^{d_{model} \times d_k}$, $W_i^K \in R^{d_{model} \times d_k}$, $W_i^V \in R^{d_{model} \times d_v}$ and $W^O \in R^{h d_v \times d_{model}}$. To sum up, the original data dimensions for a single person were (20, 9, 64, 1000), representing trials, frequency bands, EEG channels, and EEG data containing 4 s sampled at 250 Hz. The dimensions of the EEG data are transformed to (12, 40) after passing through our transformer layer. Additionally, the 60-dimensional eye tracking data was combined and expanded to dimensions (540, 1), and then fed into the fully connected layer. It's worth noting that the imbalance in the size of input dimensions between different modes does not adversely influence the classification result [40].

2.4.2. Training and Testing

In deep learning fusion, our baseline comparison comprises depres-

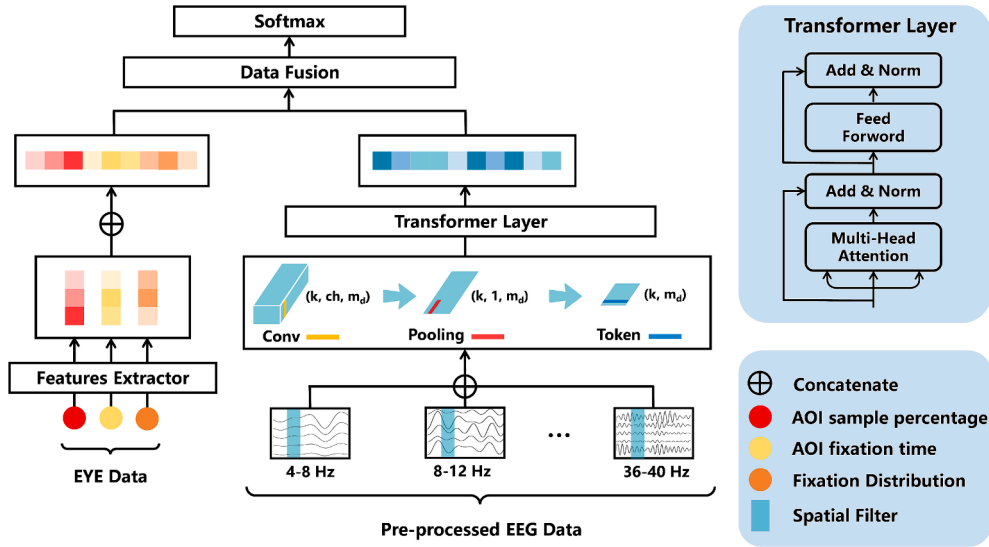


Fig. 2. The overall structure of MTNet. k represents the number of kernels, which is set at 40; ch refers to the processed EEG data channel, which is 288, and m_d denotes the processed EEG data points, which are dynamically adjusted through the convolution operation.

sion detection models including GICN [27], EEG Conformer [47], Transformer [28], CNN [48], MSGFN [49], EEGNet [50], and GAT [51]. To ensure a thorough performance evaluation, we employed multiple evaluation metrics, including accuracy, F1-score, sensitivity, and specificity. The model training process involved using the balanced cross-entropy loss function and the Adam optimization algorithm, implemented on an NVIDIA RTX 2060 GPU. We utilized Python 3.8, within the PyTorch framework, to carry out the training. The learning rate, batch size, and dropout rate are set to 0.001, 4, and 0.5, respectively. The number of execution times (N) is set to 4, and the number of heads (h) is set to 8. In our experiment, we implemented stratified three-fold cross-validation, ensuring that each fold maintains the same sample distribution ratio as the entire dataset. This approach also secured the independence of the training and test sets during the network model training.

2.5. Fusion stage

In our study, we conducted an experiment on the fusion stage in deep

learning fusion. Specifically, we analyzed the impact of early, intermediate and late fusion on the experimental results, as illustrated in Fig. 3.

- 1) Early fusion: In feature-based fusion, we performed feature extraction on the EEG and eye tracking data, thus avoiding the direct fusion of raw data and mitigating the impact of varying sampling rates. Utilizing the extracted multimodal features, we employed the fully connected neural network (FCNN) to evaluate the classification performance of early fusion. Given the fixed dimension of the input features, our focus was primarily on the number of network nodes in the first layer.
- 2) Intermediate fusion: Deep learning architectures are well-suited for intermediate fusion, typically utilizing feature vectors. Building upon our proposed MTNet, we performed an investigation into intermediate fusion. We experimented with three parameter sets to optimize the classification performance by adjusting the number of nodes in the middle layer.
- 3) Late fusion: This approach involves establishing submodels for eye tracking and EEG modes, making predictions in equal or weighted

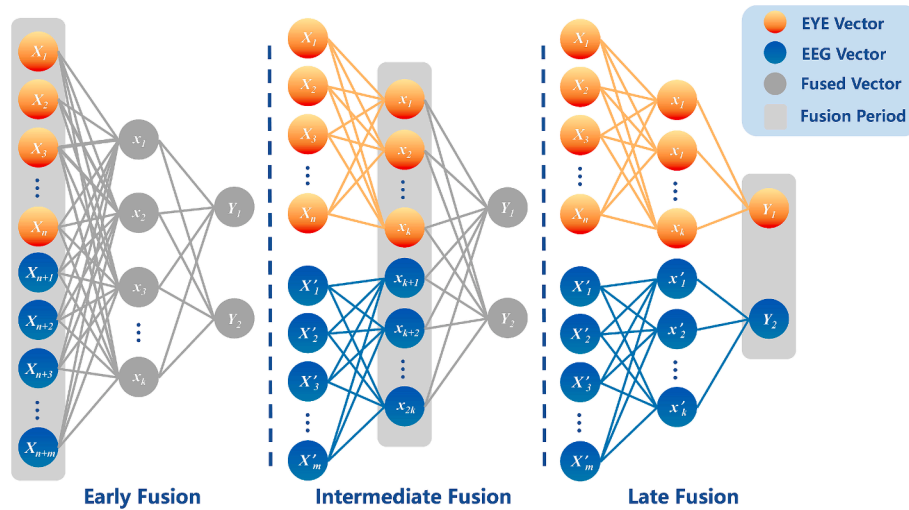


Fig. 3. The number of network layers in different stages is represented by a simplified single-layer network for visualization. Here, X denotes the input vector, n represents the number of eye tracking vectors, m denotes the number of EEG vectors, k and k' represent the number of output nodes of the middle layer network. Y_1 and Y_2 denote the output nodes of the binary classification network.

manners [40]. We utilized the FCNN for eye tracking data and the transformer model for EEG data, generating prediction categories. We conducted experiments employing two weighting methods: average weighting and entropy weighting. In our study, we selected a weight ratio of one to eight by considering the extraction dimensions of eye tracking and EEG data in deep learning fusion.

3. Results

We conducted experiments based on the fusion strategy and fusion stage for multimodal fusion of the EEG and eye tracking data. Within the deep learning fusion, we investigate three fusion stages, including early, intermediate, and late fusion, aiming to achieve optimal classification performance.

3.1. The classification results of feature-based fusion

We performed three classification tasks: eye tracking data, EEG data, and multimodal fusion data. Adaboost achieved the highest accuracy of 75.61 % when using eye tracking data. For EEG data, SVM yielded the best results with an accuracy of 81.62 %. In feature-based fusion, we combined the feature vectors of eye tracking and EEG data to obtain a feature vector dimension of 260, and SVM achieved the highest accuracy of 83.70 %. These classification results are summarized in Table 1. Feature-based fusion improved classification performance in all methods including random forest, SVM, adaboost, decision tree, Naive Bayes and KNN, compared to using eye tracking or EEG data alone.

3.2. The classification results of deep learning fusion based on different fusion stages

In our deep learning fusion classification task, we conducted experiments on EEG data and multimodal fusion. Specifically, when it comes to the classification of EEG data, the transformer model achieved the highest classification accuracy of 87.87 %. EEG Conformer, MSGFN, CNN, EEGNet, GAT and GICN reached 81.74 %, 83.58 %, 83.70 %, 79.78 %, 85.66 % and 85.54 % respectively. The results are presented in Table 2. The transformer model demonstrates the most promising result with an F1 score of 0.904.

According to the aforementioned results, it is evident that the transformer model excels in processing EEG data and attains superior performance in terms of accuracy, sensitivity, and F1-score. We performed three different fusion stages for the integration of eye tracking and EEG data. In early fusion, with a fusion network middle layer of 16 nodes, adjusting the input layer to 256 nodes yielded the highest classification accuracy of 85.91 %, accompanied by a sensitivity of 85.00 %, specificity of 82.76 %, and F1-score of 0.837. Based on our proposed MTNet, eye tracking and EEG vector are separated as inputs in intermediate fusion to obtain the optimal configuration of 128 nodes in the middle layer, with a classification accuracy of 91.79 %, sensitivity of 90.48 %, specificity of 92.86 %, and an f1 score of 0.904. When it comes to late fusion, the average weighted demonstrated superior performance, achieving a classification accuracy of 89.71 %, sensitivity of 90.00 %, specificity of 89.66 %, and an f1 score of 0.878. The results are shown in Table 3.

Table 1

Performance of the feature-based fusion based on manual feature extraction.

Classifiers	ET	EEG	Fusion
Random Forest	69.36 %	73.41 %	77.82 %
SVM	73.28 %	81.62 %	83.70 %
Naive Bayes	67.16 %	75.74 %	77.70 %
KNN	71.32 %	77.57 %	79.66 %
Adaboost	75.61 %	79.53 %	81.74 %
Decision Tree	69.24 %	71.45 %	73.53 %

Table 2

Performance of the deep learning fusion based on EEG data and fusion of eye tracking and EEG data. EEG Conformer, CNN, MSGFN, EEGNet, GAT and GICN were assessed using EEG data, with the Transformer method utilizing the MTNet framework for processing EEG independently. The intermediate fusion involved the use of the proposed MTNet.

Model	Accuracy	sensitivity	specificity	F1-score
EEG Conformer [47]	81.74 %	77.27 %	85.19 %	0.791
MSGFN [49]	83.58 %	84.21 %	83.33 %	0.800
CNN [48]	83.70 %	80.95 %	85.71 %	0.810
EEGNet [50]	79.78 %	76.19 %	82.14 %	0.762
GAT [51]	85.66 %	85.00 %	86.21 %	0.829
GICN [27]	85.54 %	88.89 %	83.87 %	0.821
Transformer [28]	87.87 %	85.71 %	89.28 %	0.857
Ours (Early fusion)	85.91 %	85.00 %	82.76 %	0.837
Ours (Intermediate fusion)	91.79 %	90.48 %	92.86 %	0.904
Ours (Late fusion)	89.71 %	90.00 %	89.66 %	0.878

Table 3

Performance of the early fusion, intermediate fusion, and late fusion on the experimental results.

Fusion stage	Hyper parameters	Selection range	Accuracy
Early fusion	Network node	[64, 8, 2]	83.67 %
		[128, 16, 2]	81.74 %
		[256, 16, 2]	85.91 %
Intermediate fusion	Network node	[64, 2]	87.87 %
		[128, 2]	91.79 %
		[256, 2]	89.83 %
Late fusion	Weighted method	Average Weighted	89.71 %
		Entropy Weighted	85.66 %

3.3. Visualization and comparison

To effectively present our experimental results, we employed column chart and confusion matrices for visualization. In the column chart, we employed accuracy and F1-score as the evaluation metrics. The confusion matrix is a tool used to assess the performance of a classification model. In this matrix, each row represents samples of the actual class, while each column represents samples predicted by the model. Fig. 4 and Fig. 5 provide all the comparisons, with the feature-based fusion representing the SVM classifier's outcomes. In Table 4, MTNet is compared with other methods, including individual modes and the results post fusion [52–54]. The findings indicate that the proposed network effectively leverages the spatiotemporal characteristics of multimodal information.

4. Discussion

In response to the rising rates of depression, our study introduces an objective, comprehensive, and accurate detection methodology. A multimodal signal acquisition platform is established to delve into the visual attention mechanism and cognitive brain activity using non-invasive eye tracking and EEG technologies. Subsequently, we have introduced a novel multimodal transformer network (MTNet) to extract both local and global features from the multimodal data, resulting in a classification accuracy of 91.79 %, surpassing machine learning algorithms and baseline models. In addition to the fusion strategy, the fusion stage also demonstrated a crucial role in our experiments. Intermediate fusion demonstrates superior feature extraction capability, surpassing both early and late fusion, particularly in capturing interactions between modal features.

Compared to single-modal data, the fusion of multimodal data exhibits enhanced feature extraction capabilities, leading to a substantial improvement in the f1 score, which reflects the overall classification performance of the model. This fusion approach also mitigates the impact of sample imbalance on classification outcomes. In our study, the

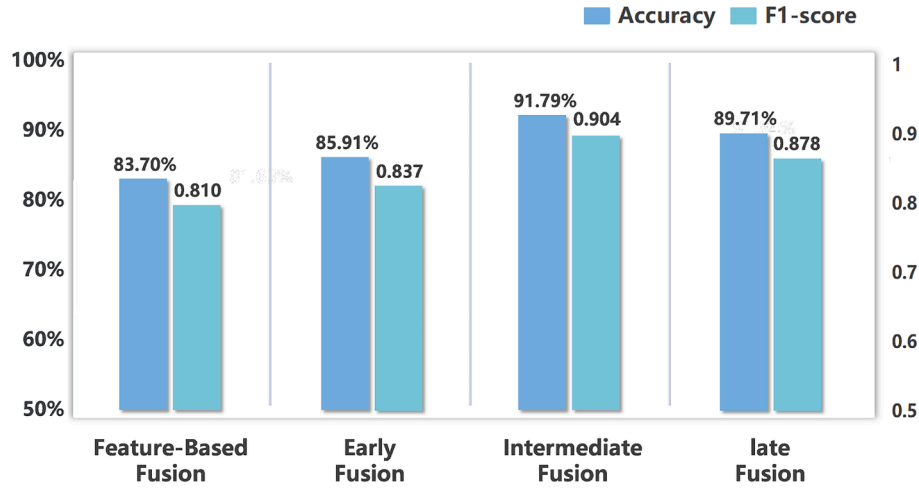


Fig. 4. The bar chart illustrates a comparison between the performance of feature-based fusion and deep learning fusion methods. Accuracy metrics are represented on the left vertical axis, while F1 scores are depicted on the right vertical axis.

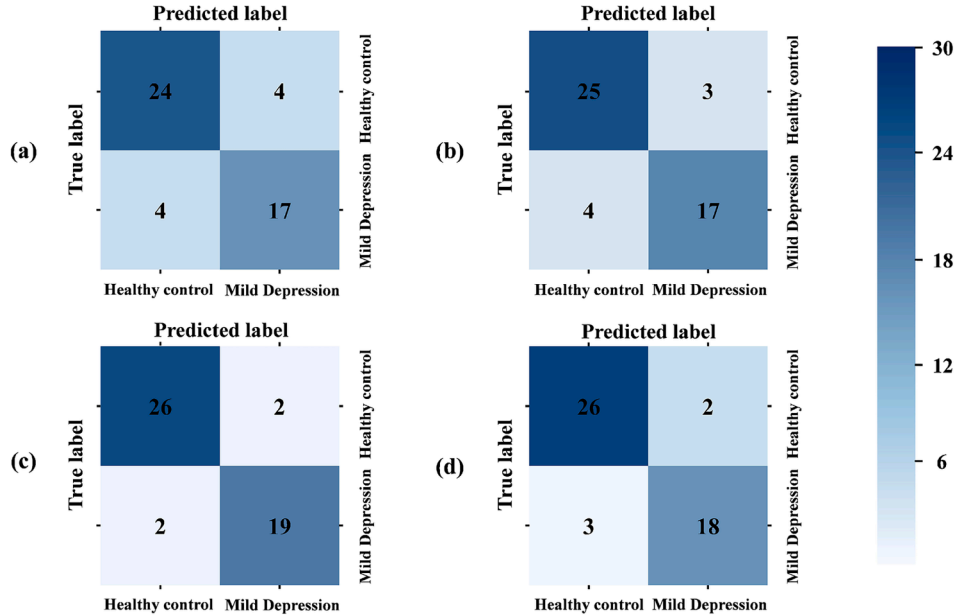


Fig. 5. The confusion matrix based on classification results from (a) feature-based fusion, (b) early fusion, (c) intermediate fusion, and (d) late fusion.

Table 4

MTNet was compared with the results of other multimodal classification tasks that combined eye tracking and EEG, specifically in the context of depression and emotion recognition. These tasks involved the multimodal analysis of eye tracking and EEG.

References	Classifier	ET	EEG	Fusion
Ding et al. [36]	Logistic Regression	65.30 %	72.41 %	79.63 %
Zhu et al. [52]	DBN	78.72 %	85.83 %	89.54 %
Guo et al. [53]	DNN	71.99 %	73.98 %	79.63 %
Zhu et al. [54]	Bayesian Rule	74.80 %	84.27 %	91.12 %
Ours (MTNet)	Transformer	75.61 %	87.87 %	91.79 %

F1-scores demonstrated a notable enhancement, rising from the highest single mode score of 0.857 to 0.904 upon implementing the MTNet algorithm. This improvement aligns with findings from various prior studies on depression detection [52,53,55]. Zhu et al. [52] employed physiological signals to identify depression, achieving an accuracy of 85.83 % using EEG features alone. However, by integrating EEG features

with eye tracking data, the accuracy significantly increased to 89.54 % using deep belief network (DBN), indicating an improvement of 3.71 %. Similarly, Guo et al. [53] utilized eye movement, and EEG data for multimodal emotion recognition. The deep neural network (DNN) achieved an accuracy of 79.63 %, surpassing the accuracy of both the eye tracking mode (71.99 %) and the EEG mode (73.98 %). Furthermore, Wu et al. [55] proposed a deep gradient network for emotion recognition, which fused eye tracking and EEG data. This fusion approach resulted in a classification accuracy of 88.10 %.

Our research supports the superiority of deep learning fusion over feature-based fusion for effectively utilizing multimodal data. In feature-based fusion, SVM demonstrate the highest classification performance in multimodal data, achieving an accuracy of 83.70 %. This accuracy is higher than of the single mode data, highlighting its stronger feature extraction capabilities compared to the other machine learning classifiers. Besides, our proposed multimodal network, based on transformer model, leverages its global receptive field to achieve proficiency in extracting spatiotemporal characteristics. As a result, we achieved an accuracy of 91.79 %, surpassing the performance of feature-based fusion

by 8.09 %. In the classification task of deep learning algorithms applied to EEG data, it is noteworthy that the transformer-based classifier achieved a classification accuracy of 87.87 %, surpassing the performance of feature-based fusion relying on multimodal data. Zhu et al. [54] integrated EEG and eye tracking data for motor imagination tasks, achieving an 82.86 % classification accuracy using decision layer fusion with D-S evidence theory, outperforming the feature layer by 1.7 %.

In addition to the aforementioned fusion strategy, the fusion stage plays a crucial role in data fusion. Our study revealed that intermediate fusion yields superior performance compared to early fusion and late fusion. In our experiment, we conducted early, intermediate, and late fusion, respectively, by adjusting the parameter ranges or classifiers based on different fusion stages. In early fusion, we combined the features of the eye tracking and EEG data in the initial layer without distinguishing their sources. However, intermediate fusion, leveraging prior knowledge of the mode the features originate from, effectively accommodated the heterogeneity of multimodal data compared to early fusion. On the other hand, late fusion showed limitations in capturing the interaction between different modal features. This drawback is expected to have a minimal impact on results in scenarios with low pattern correlation.

Our study demonstrates that early fusion exhibit lower classification performance, specificity, and F1-score compared to intermediate fusion and late fusion. This observation suggests that early fusion may not adequately extract features, potentially limited by the varying sampling rates in different modes. Hence, when dealing with the multimodal fusion of biological signals, early fusion does not exhibit clear advantages over intermediate fusion and late fusion. Furthermore, we discovered that the sensitivity of depression detection could be significantly enhanced through multimodal data fusion. Specifically, intermediate achieved a sensitivity of 90.48 %, while late fusion achieved a sensitivity of 90.00 %. This improvement could prove invaluable in preventing misdiagnosis during the diagnostic process.

Biomedical applications encounter specific challenges when it comes to multimodal fusion, such as small combined dimensions relative to sample sizes, incomplete modes, and dimensional imbalances between modes [34,35]. By constructing a multimodal synchronous acquisition platform, our classification task has integrated biological signal data, achieving a classification accuracy rate exceeding 90 %. To further augment the applicability of our multimodal acquisition platform and enhance algorithmic classification performance, we are planning to increase the number of participants, thereby expanding the dataset size. Moreover, we aim to leverage our synchronous collection platform to encompass a wider spectrum of diseases, including depression and bipolar disorder, within our diagnostic classification.

5. Conclusion

In our research, we developed a multimodal acquisition platform to explore feature-based fusion and deep learning fusion for EEG and eye tracking data. Our proposed MTNet based on transformer structure, comprehensively extracts the spatiotemporal features of multimodal data, and the classification accuracy reaches 91.79 %, surpassing the feature-based fusion. Moreover, when considering the fusion stage, we examined three stages: early, intermediate, and late fusion. Our study highlights the superiority of intermediate fusion over early fusion and late fusion when processing EEG and eye tracking data due to its exceptional adaptability to the heterogeneity of multimodal data.

CRediT authorship contribution statement

Feiyu Zhu: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Jing Zhang:** Resources. **Ruochen Dang:** Writing – review & editing. **Bingliang Hu:** Supervision. **Quan Wang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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